Networks and the Internet

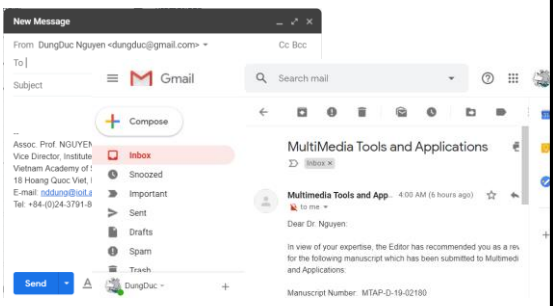
NGUYEN Duc Dung, Ph.D.
Institute of Information Technology, VAST

Informatics: Course Content

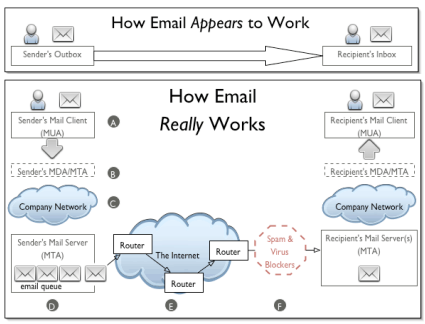
- History of Computing and Computer Architecture *(no practice)*
- Data Presentation and File Structures
- **Networks and the Internet**
- Operating System
- Software Tools for Techies
- Introduction to Programming and Software Engineering

Your Everyday Applications?

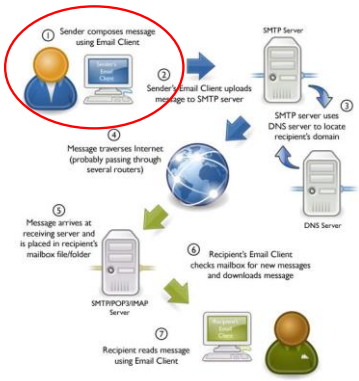
How to Email?



How Email Works?

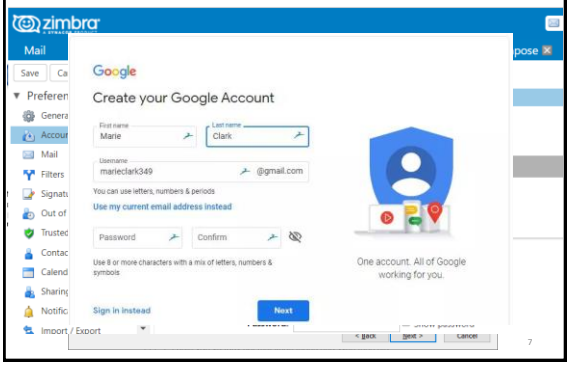


How Email Works?

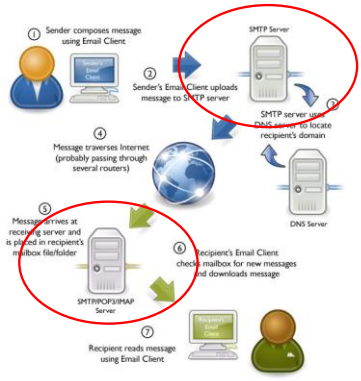


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Email Client

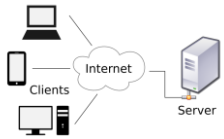


How Email Works?



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Server: SMTP vs. SMTP/POP/IMAP,…



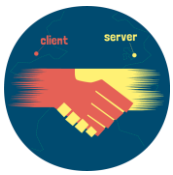
- SMTP: Simple Mail Transfer Protocol
- POP: Post Office Protocol
- IMAP: Internet Message Access Protocol
- FTP: File Transfer Protocol
- …

Internet Protocols

Application layer
BGP DHCP DNS FTP HTTP HTTPS IMAP LDAP MGCP MQTT NNTP NTP POP ONC/RPC RTP RTSP RIP SIP SMTP SN Telnet TLS/SSL XMPP *more…*
Transport layer
TCP UDP DCCP SCTP RSVP QUIC *more…*
Internet layer
IP IPv4 IPv6 ICMP ICMPv6 ECN IGMP Ipsec *more…*
Link layer
ARP NDP OSPF Tunnels L2TP PPP MAC Ethernet Wi-Fi DSL ISDN FDDI *more…*

?

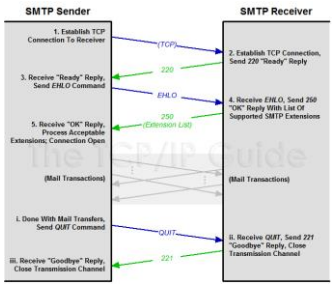
Communication Protocols: What?



- In telecommunication, a **communication protocol** is a **system of rules** that allow two or more entities of a communications system to transmit information via any kind of variation of a physical quantity.

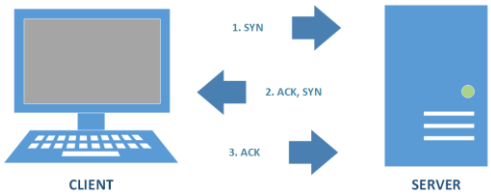
https://en.wikipedia.org/wiki/Internet_Protocol

SMTP



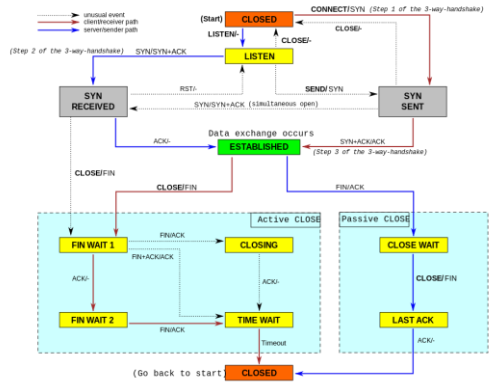
http://www.tcpipguide.com

TCP/IP Three-way Handshake



TCP Three-Way Handshake

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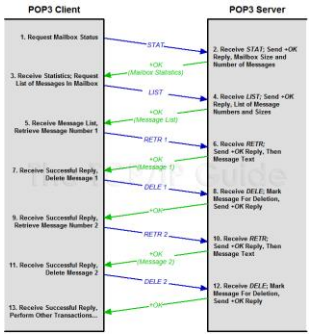
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Table 4-4, Timing diagram for a communication protocol

time period	computer 1	computer 2
1	Listening	Listening
2	Are you ready?	
3		Yes, I am
4	Here comes part 1	
5		I received part 1
6	Here comes part 2	
7		I received part 2
8	I'm finished	
9		Terminate

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POP3



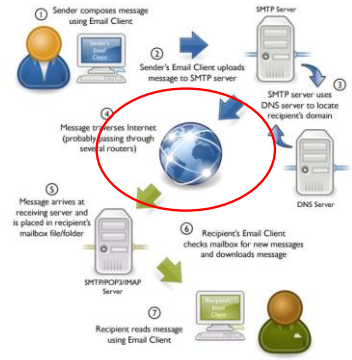
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Communication Protocols: Why?

- **SMTP** (Simple Mail Trans. Protocol): Protocol used in **sending e-mail**.
- **POP3** (Post Office Protocol V3): Protocol for **receiving e-mail** from a mail server
- **IMAP:** A standard protocol for **accessing e-mail** from a mail server.
- **HTTP** (Hypertext Transfer Protocol) – A protocol designed for **transferring files** (primarily content files) on the World Wide Web.
- **FTP** (File Transfer Protocol) – A protocol designed to **exchange text and binary files** via the Internet
- …

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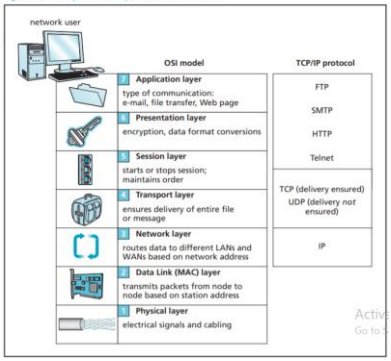
How Email Works?



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Figure 5-4, TCP/IP protocols compared with the OSI model

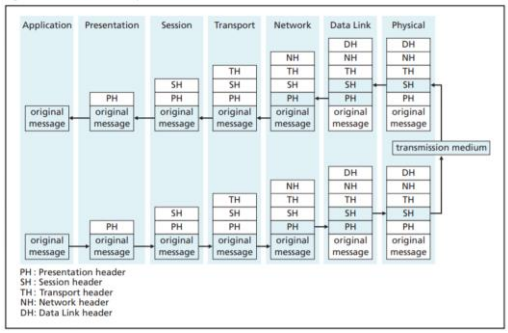


OSI:
Open System Interconnection

TCP/IP:
Transmission Control Protocol / Internet Protocol

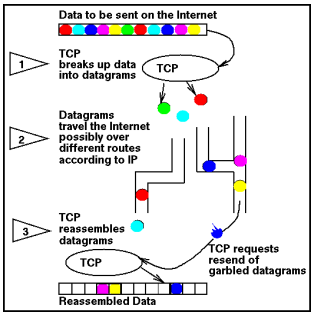
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Figure 4-3, How the OSI model processes data



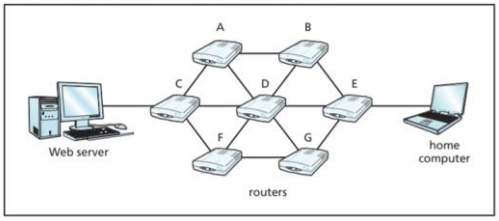
20

Data Transmission on the Internet



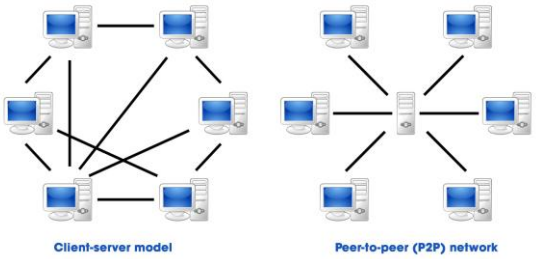
21

Figure 5-3, Routers provide many alternative routes for packets



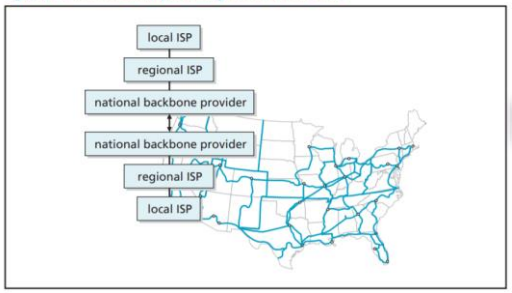
22

P2P vs. Client Server



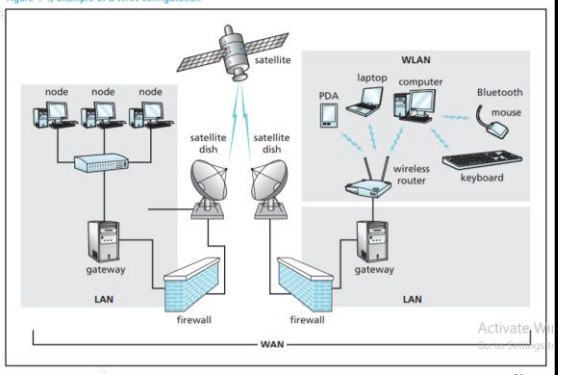
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Figure 5-1, Internet data can pass through several levels of ISPs



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Figure 4-4, Example of a WAN configuration



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How Email Works?

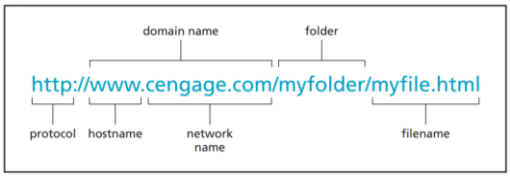


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URL and DNS

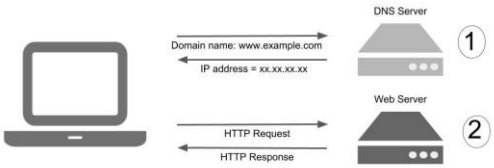
Figure 5-6, Structure of a URL



- **URL:** Uniform Resource Allocator
- **DNS:** Domain Name System

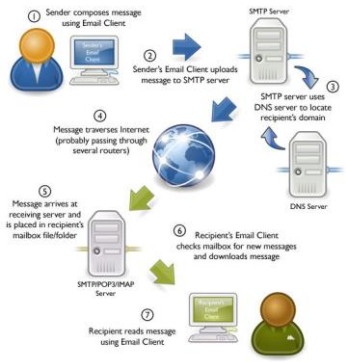
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DNS Server



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How Email Works?

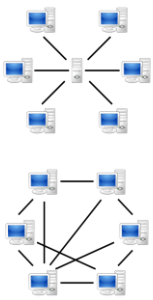


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Internet Applications

- E-Mail
- Web browsing
- Instant messaging
- Social networking
- Search engine
- Media sharing
- Peer-to-peer networking
- …



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HTML and Web Browser

HTML

- HTML (Hypertext Markup Language) is the standard markup **language** for Web pages (content).
- Example:

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>
<h1>This is a Heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

Web Browser

- Software for **accessing information** on the Web (view).
- Example:

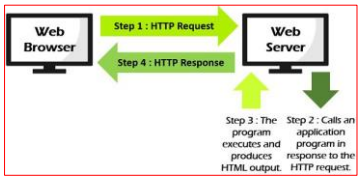


This is a paragraph.

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Static vs. Dynamic Web Pages





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Homework

- Create a Gmail account
- Create cloud driver, upload and share images
  - Google
  - Dropbox
- Create HTML document with
  - Text
  - Image
  - Video/Audio

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